Same Deadline, Different Persistence

Group 9: Jiayang Liu, Yanbin Chen, Yanshun Wang, Yuming Zhang

Data Mining Course Project

Question

Within the same module presentation, among students whose clicks after the first TMA deadline are comparable, are submission timing and post-deadline persistence associated with failing rather than passing?

Design

Anchor: the deadline of the first positive-weight TMA. Windows: 14 days before and 14 days after. Fail is the positive class; Withdrawn is described separately. Pairs: 1:1 within presentation on post-deadline clicks, 5% tolerance, zero clicks only with zero clicks.

Evidence

Cohort: 24,125 eligible records, 19,806 in the binary comparison, 6,092 Fail. Paired differences (Fail − Pass): submitted_by_cutoff: -0.124; submitted_after_due: +0.084; relative_submission_day: +1.381; post_active_days: -0.368; post_longest_inactive_run: +0.389; post_week2_share: -0.037. Model M2 Fail AP 0.661 against a no-skill floor of 0.308.

Reading

The pairing conditions on post-deadline volume, so it cannot estimate the overall failure rate. Persistence separates outcomes descriptively far more clearly than it improves prediction, and the score model is a retrospective upper bound.

LaTeX TikZposter